

RAS and CAS, the memory didn't come up. The processor tried *talking* to the memory, but it always read back zeroes.

It sounds very simple at this point, but considering the fact that nothing else was working at that point, isolating the issue to its true cause was quite a challenge—given that it was unknown whether the layout was done improperly or there were issues with the PCB, impedance matching and so on.

To find a solution, we started off by first checking whether all the signals were okay or not. We began by using a megahertz oscilloscope to make sure that every signal works out. Then we had to go through the DVR controller, which had 25 registers, with each register having around ten parameters. So there were a total of 250 various bit settings to be checked.

Even after doing all this, the message displayed was 'Ok.' So we went back to the papers and did one more review. That's where one of our very experienced designers spotted this particular flip.

The second problem was how to fix it. Pins were located pretty close to each other, and we had to cross two of those. So we took an extremely thin wire, connected it in and pulled off a criss-cross right there. Once we did this, the tablet booted up to Android.

Between resistive and capacitive types, which screen did you choose and why?

We chose a capacitive screen because resistive screens require a certain amount of pressure to work. The pressure causes a certain amount of resistance, and that is how the screen works.

Capacitive screens, on the other hand, work by sensing a distortion in the field. When you place a finger on the screen, it creates a distortion in the electrostatic field, which is measured internally.

Now if a protective glass on top is desired by the user, a resistive screen will not work as it is not possible to press it through the thick glass. Whereas, a capacitive screen will still sense distortion in the field because glass is a dielectric.

The capability to improve the strength of the tablet by placing glass on the display while still allowing touch-screen functionality is what made us select a capacitive screen.

However, there is one particular scenario where resistive screens would score over capacitive screens: When the humidity is high, such as when it is raining or you are in a high humidity area, capacitive screens lose their response. This is because humidity makes the field distortion difficult to detect. In such cases, resistive touchscreens perform better.

Another example where resistive screens are better would be when you are wearing gloves, such as in an industrial environment.

You are targeting systems integrators, aka solution providers, for STAMP tablets—why?

While the tablet is not basically different, the needs of a solution provider are different from the needs of a consumer. Typically, consumers buy tablets for enjoying multimedia, gaming, reading e-books and browsing the Internet. Then there are people who want to provide solutions based on tablets—menu cards on tablets, home control panels based on tablets and so on. They need to run a very specific application, which means they do not need all the frills and features of a consumer tablet. What they really need is a relatively simple and more reliable tablet that can be used in a robust environment.

We are catering to the needs of these users. For instance, if a tablet is to be used for home control or appliance control, hardware and interface need to be added to it. On the other hand, if it is to be used in retail or e-commerce application, a credit card swiper, thermal printer or bar-code scanner needs to be included depending on the requirements. This level of customisation and support is not easy to find in imported tablets.

By producing a tablet that is fully developed by us, we have total control over it. We can do a lot of customisation so that it fits the clients' needs. ●

(The complete interview can be read at electronicsforu.com)

EFY Group's New Initiative

EB TIMES Launched

This 8-page monthly **B2B Newspaper** will be a resource for traders, distributors, dealers, and those who head channel business, as it aims to give an impetus to channel sales

